Experiment No-20 : Planets Database

Aim:

To write a Prolog program to create a PLANETS database and answer queries about the planets of the solar system.

Programming Code:

```
% planet(Name, Order, DiameterKm, DistanceFromSunMillionKm, Type)
planet(mercury, 1,   4879,   57.9, terrestrial).
planet(venus,   2,  12104,  108.2, terrestrial).
planet(earth,   3,  12756,  149.6, terrestrial).
planet(mars,    4,   6792,  227.9, terrestrial).
planet(jupiter, 5, 142984,  778.5, gas_giant).
planet(saturn,  6, 120536, 1432.0, gas_giant).
planet(uranus,  7,  51118, 2867.0, ice_giant).
planet(neptune, 8,  49528, 4515.0, ice_giant).

% has_rings(Planet)
has_rings(jupiter).
has_rings(saturn).
has_rings(uranus).
has_rings(neptune).

% ---------- Rules ----------
% Planet at a given position from the Sun
planet_at(Order, Name) :- planet(Name, Order, _, _, _).

% Planets of a given type
planet_type(Name, Type) :- planet(Name, _, _, _, Type).

% Inner planets (first four) and outer planets
inner_planet(P) :- planet(P, O, _, _, _), O =< 4.
outer_planet(P) :- planet(P, O, _, _, _), O > 4.

% Planet A is bigger than planet B
bigger_than(A, B) :- planet(A, _, DA, _, _), planet(B, _, DB, _, _), DA > DB.

% Largest and smallest planet
largest_planet(P)  :- planet(P, _, D, _, _), \+ (planet(_, _, D2, _, _), D2 > D).
smallest_planet(P) :- planet(P, _, D, _, _), \+ (planet(_, _, D2, _, _), D2 < D).

% Planets farther from the Sun than a given planet
farther_than(P, Ref) :-
    planet(P, _, _, D1, _), planet(Ref, _, _, D2, _), D1 > D2.
```

OUTPUT:

```
$ swipl -q exp20.pl

?- planet_at(4, P).
P = mars.

?- planet_type(P, terrestrial).
P = mercury ;
P = venus ;
P = earth ;
P = mars.

?- inner_planet(P).
P = mercury ;
P = venus ;
P = earth ;
P = mars.

?- has_rings(P).
P = jupiter ;
P = saturn ;
P = uranus ;
P = neptune.

?- largest_planet(P).
P = jupiter.

?- smallest_planet(P).
P = mercury.

?- bigger_than(earth, venus).
true.

?- farther_than(P, jupiter).
P = saturn ;
P = uranus ;
P = neptune.

$
```

Result

The PLANETS database was implemented successfully using Prolog. The program stores the details of the eight planets and the queries correctly return planets by position, type, size and distance from the Sun.